1. 1^3 + 2^3 + … + n^3 = ((n+1)(n+2)/2)^2

<u>Basis</u>: First value is 1, so n = 1.
n^3 = ((n(n + 1)/2))^3
1^3 = 1
((1(1+1)/2))^3 = 1
Therefore P(1) is true.

<u>Inductive Step</u>: Assume true for n = k by IH.
Since P(k) is true, now must prove P(k + 1)
1^3 + 2^3 + … + k^3 + (k+1)^3 = ((k+1)(k+2)/2)^2
Left side: ((k+1)(k+2)/2)^2 + (k+1)^3
(k^2(k+1)^2/4) + (k+1)^3
(k^2(k+1)^2/4) + (4(k+1)^3)/4
Right side: ((k+1)(k+2)/2)^2
((k+1)(k+2))^2/4

<u>Simplification:</u>
(k^2(k+1)^2/4) + (4(k+1)^3)/4 = ((k+1)(k+2))^2/4
- Remove the division by 4.
(k^2(k+1)^2 + (4(k+1)^3) = ((k+1)(k+2))^2
- Split (k+1)^3 into (k+1)^2(k+1)
(k^2(k+1)^2 + 4(k+1)(k+1)^2 = ((k+1)(k+2))^2
- Factor out (k + 1)^2 and simplify 4(k + 1)
(k+1)^2(k^2 + 4k + 4) = ((k+1)(k+2))^2
- Simplify into (k+2)^2
(k+1)^2(k+2)^2 = ((k+1)(k+2))^2
Both sides are equivalent, therefore P(k+1) is true.
k = n, therefore P(n + 1) is also true.
Since P(1) is true and P(n+1) is also true as shown above, the statement is true for all positive integers n in P(n).

2. $1 \cdot 1! + 2 \cdot 2! + \cdot \cdot \cdot + n \cdot n! = (n + 1)! - 1$, for all positive integers n.

<u>Basis:</u> First value is 1, so $n = 1$
$n \cdot n! = (n + 1)! - 1$
$1 \cdot 1! = 1 \mid (1+1)! - 1 = 2 - 1 = 1$
$1 = 1$, therefore P(1) is true.

<u>Inductive Step:</u> Assume true for $n = k$ by IH.
Since P(k) is true, now must prove P(k + 1)
Now add $(k + 1) \cdot (k + 1)!$ to both sides and plug in (k+1) for k.
$1 \cdot 1! + 2 \cdot 2! + \cdot \cdot \cdot + k \cdot k! + (k + 1) \cdot (k + 1)! = ((k+1) + 1)! - 1 + k \cdot k! + (k + 1) \cdot (k + 1)!$
Left Side: $(k + 1)! - 1 + (k + 1)*(k+1)! \rightarrow (k+1)(k)! - 1 + (k + 1)(k+1)(k)!$
$(k+1)!(k + 1 + 1)) - 1 \rightarrow (k+1)!(k + 2)) - 1 \rightarrow (k + 2)! - 1$
Right Side: $((k + 1) + 1)! - 1$
$(k + 2)! - 1$
Together:
$(k + 2)! - 1 = (k + 2)! - 1$
$n = k$, therefore P(n + 1) is therefore true.
By principle of MI, since P(1) is true and P(n + 1) is also true, statement is true for all P(n) when n is positive integer.

3. $2 - 2 \cdot 7 + 2 \cdot 7^2 - \cdots + 2(-7)^n = (1 - (-7)^{(n+1)})/4$, for all non-negative integers n.

<u>Basis:</u> First value is 0, therefore we prove P(0)
$2(-7)^0 = 2$
$(1 - (-7)^{(0+1)})/4 = 2$
2 = 2, therefore P(0) is true.

<u>Induction Step:</u> Assume true for n = k by IH.
$2 - 2 \cdot 7 + 2 \cdot 7^2 - \cdots + 2(-7)^k = (1 - (-7)^{(k+1)})/4$
Prove P(k + 1) is also true, also known as n = k + 1
$2 - 2 \cdot 7 + 2 \cdot 7^2 - \cdots + 2(-7)^k + 2(-7)^{(k+1)} = (1 - (-7)^{(k+1)})/4 + 2(-7)^{(k+1)}$
$(1-(-7)^{(k+1)})/4 + 2(-7)^{(k+1)}$
$(1 - (-7)^{(k+1)}) + 8(-7)^{(k+1)})/4$
$(1 + 7(-7)^{(k+1)})/4$
$(1 - (-7)(-7)^{(k+1)})/4 \rightarrow (1 - (-7)^{(k+2)})/4$
$\mathbf{(1 - (-7)^{(k+2)})/4 = (1 - (-7)^{(k + 1 +1)})/4}$
n = k, therefore P(n + 1) is true.
Both sides are equivalent, therefore it is proven that P(n+1) is true. Since the basis is also proven for P(n), then the statement is true for all of P(n).

4. $1/(1 \cdot 2) + 1/(2 \cdot 3) + \ldots + 1/n(n+1) = n/(n+1)$ for all positive ints n.

Basis: First value is 1, therefore we prove P(1)
$1/(1)((1) + 1) = ½$
$(1)/((1)+1) = ½$
$½ = ½$, therefore P(0) is true.

Induction Step: Assume true for n = k by IH.
$1/(1 \cdot 2) + 1/(2 \cdot 3) + \ldots + 1/k(k+1) + 1/(k+1)((k+1)+1) = k/(k+1)$
$k/(k+1) + 1/(k+1)(k+2)$
$(k(k+2) + 1)/(k+1)(k+2)$
$(k^2 + 2k + 1)/(k+1)(k+2)$
$(k+1)(k+1)/(k+1)(k+2)$
$(k+1)/(k+2)$
Set equal to right side with k+1 plugged in as k.
$(k+1)/(k+2) = (k+1)/((k+1)+1) \rightarrow (k+1)/(k+2)$
Both sides are equivalent, therefore P(n+1) is proven as true. Since basis (P(n)) has also been proven as true, then the statement is true for all of P(n).

5. 6 divides $n^3 - n$, for all positive integers n.

<u>Basis:</u> First value is 0, so $n = 0$.
$0^3 - 0 = 0$ and $0/6 = 0$, therefore P(n) is true when $n = 0$.

<u>Inductive Step:</u> If P(n) is true, then prove P(n+1)
Assume P(n) is true for some n
Assume true for $n = k$ by IH.
$(k+1)^3 - (k + 1) = k^3 + 3k^2 + 3k + 1 - k - 1$
$(k^3 - k) + (3k^2 + 3k) \rightarrow (k^3 - k) + 3k(k+1)$
- Either k or k + 1 is even, so divide 3k(k+1) by 2.

$3k(k+1)/2 \rightarrow k(k+1)/6$
- This shows that 6 divides 3k(k+1)
- Through inductive hypothesis, 6 divides $n^3 - n$, so 6 divides sum $3k(k+1) + (k^3 - k)$
- Since $n = k$, 6 also divides $3n(n+1) + (n^3 - n)$.
- Since $3n(n+1) + (n^3 - n) = (n+1)^3 - (n+1)$, it is proven above that 6 divides $(n+1)^3 - (n+1)$, showing that $P(n) \rightarrow P(n+1)$.
- It is proved that 6 divides $n^3 - n$ for all n (pos ints).

6. Let P(n) be the statement that n-cent postage can be formed using just 4-cent and 7-cent stamps. Prove that P(n) is true for all n ≥ 18, using the steps below.

6A. First, prove P(n) by regular induction. State your basis step and inductive step clearly and prove them.

<u>Basis:</u> Prove that P(18) is true because 18 is the lowest amount of money possible.
7c + 7c + 4c = 18c, therefore P(18) is true.

<u>Inductive Step:</u> Let k >= 18
Assume: k cent postage can be formed using 4c + 7c.
Prove: (k + 1) c can be formed using 4c + 7c.
<u>Case 1:</u> Suppose at least one 4c stamp was used to form k c postage
- Adding a 4c stamp will cause the total amount of money to increase by 4c.
- Obviously, k + 4 != k + 1, but if you also replace a 7c stamp with a 4c one, k + 1 can be achieved.

Therefore, P(k+1) is true in this case.
<u>Case 2:</u> There are no 4c stamps used to form k cents postage.
Only 7c stamps are used, implying k is multiple of 7. Since k >= 18, follows that k >= 21 since 21 is the smallest multiple of 7 greater than 18. At least 3 7c stamps are used. Replace the stamps with 2 7c stamps and two 4c stamps and it results in k + 1 cents, as k - 1(7) + 2(4) = k + 1
Therefore, P(k+1) is proven in this case.
In both cases, P(k + 1) is proven. Therefore by regular induction, P(n) is proven for all n (pos ints over 17).

6B. Now, prove P(n) by strong induction. Again, state and prove your basis step and inductive step. Your basis step should have multiple cases.
<u>Basis:</u>
18 cents = 7c + 7c + 4c
19 cents = 7c + 4c + 4c + 4c
20 cents = 4c + 4c + 4c + 4c + 4c
21 cents = 7c + 7c + 7c

<u>Inductive Step:</u> Let k >= 21. We assume that for all l, where 18 <= l <= k, postage of L cents can be formed using 4c + 7c
- Prove that k + 1 cents postage can be formed.
- Since k >= 21, therefore k - 3 >= 18, so k - 3 cents is postage possible (by IH)
- Add a 4c stamp to k - 3 cents postage to form k + 1 cents postage, since (k-3) + 4 = k + 1
Therefore, P(k + 1) is true and P(k) is true for all k (above 17 and positive integer)

7. Suppose $P(n)$ is true for an infinite number of positive integers n. Also, suppose that $P(k + 1) \rightarrow P(k)$ for all positive integers k. Now, prove that $P(n)$ is true for all positive integers. This is the reverse induction principle.

P1: $P(n)$ is true for infinite amount of pos ints n.
P2: $P(k + 1) \rightarrow P(k)$ for all pos ints k.
Firstly, assume true for $n = k$ by IH.
Contradiction Assumption: There is some value of n where $P(n)$ is not true or doesn't compute.
P2 Contrapositive: $\neg P(k) \rightarrow \neg P(k + 1)$ for all pos ints k.
By MI, $\neg P(n)$ is true for all n, therefore $P(n)$ is not true for all n.
CONTRADICTION: By P1, it is given that $P(n)$ is true for an infinite amount of positive integer. This means that the assumption is incorrect and that $P(n)$ is true for all positive integers.